

Abhijith Premraj

Computer Vision & ML Engineer | End-to-End AI Systems

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SUMMARY

AI/ML Engineer with hands-on experience designing, optimizing, and deploying production-grade Computer Vision and Generative AI systems. Specialized in real-time model deployment, camera calibration, and pose estimation. Proven ability to take ML systems from data to production using Python, PyTorch, TensorFlow, FastAPI, and Docker.

TECHNICAL SKILLS

Programming Languages: Python (expert), Java, SQL, MATLAB, JavaScript

Deep Learning: TensorFlow, PyTorch, TFLite, CNNs, RNNs, Transformers, ViT, LLMs, Model Quantization, Transfer Learning.

Computer Vision: OpenCV, Open3D, Camera Calibration, Pose Estimation, Object Detection (YOLO), Segmentation, 3D Reconstruction

LLM/Gen AI: RAG Pipelines, LangChain, LangGraph, Agentic AI, Ollama, HuggingFace, Vector Search (Qdrant, Chroma)

DevOps/MLOps: Docker, Kubernetes, Git, MLFlow, DVC, FastAPI, Flask, Streamlit, Github Actions, Azure, Linux

EXPERIENCE

AI Engineer

Freelance

Dec 2025 – Present

Abu Dhabi, UAE

- Designed and delivered a fully air-gapped Retrieval-Augmented Generation (RAG) system for a client with strict data privacy requirements
- Ran a local LLM (Qwen via Ollama) with FAISS vector search and BM25 hybrid retrieval with no cloud API calls or internet dependency
- Implemented BGE cross-encoder reranking, PII redaction (names, emails, phone numbers, etc.), and conversation history via Redis with auto-summarization.
- Built a React, TypeScript frontend with FastAPI streaming backend (SSE), supporting document ingestion, semantic search, and chat.
- Enforced HuggingFace Hub offline mode at config level to guarantee no accidental web access.

Computer Vision Engineer

Carl Zeiss Meditec AG

Feb 2024 – Apr 2025

Oberkochen, Germany

- Delivered production-ready 6DoF surgical head-pose estimation system by architecting and integrating a CNN-based 2D–3D registration module, ICP point-cloud alignment, and real-time camera feed processing into a single validated pipeline.
- Designed coordinate-system transformation pipeline to bridge annotation and model coordinate frames; generated ground-truth bounding boxes via 3D model projection, eliminating manual labelling overhead across the full training dataset.
- Reduced calibration error through statistical outlier rejection and automated filtering in a custom camera calibration toolbox (OpenCV + MATLAB), which was subsequently adopted by internal R&D teams as the standard production tool.
- Accelerated train/val/test split optimisation using KDE-based distribution analysis, improving model generalisation and reducing validation-set bias across pose estimation experiments.
- Documented and presented findings at weekly research reviews, mentored junior engineers on camera calibration theory and practical implementation.

Automation Engineer (Assistant Manager)

CEAT Pvt. Ltd.

July 2019 – Mar 2020

Gujarat, India

- Analyzed time-data across R&D lab project workflows, identifying bottlenecks and inefficiencies in manual processes
- Automated R&D workflows using Python-based data acquisition software, reducing manual data handling by 80%
- Built lab inventory management system that enabled data-driven decisions on resource ordering and storage, improving lab resource utilization and reducing stock-outs.

PROJECTS

Production-Ready YOLO Head Detection System with MLOps & CI/CD
PyTorch, YOLOv8, MLflow, DVC, FastAPI, Docker, GitHub Actions, OpenCV

MLOps

- Engineered full data pipeline (ingestion → versioning with DVC → training → registry) achieving reproducible experiments at scale.
- Deployed RESTful inference API via FastAPI serving YOLOv8 head detection with batch, feature-map, and real-time webcam modes , simulating production traffic patterns.
- Automated CI/CD with GitHub Actions: unit tests, format checks, Docker build, and push to container registry on every commit.
- Tracked all experiments with MLflow + DagsHub, enabling model comparison and rollback across training runs.

MobileNet-UNet Lane Detection (On-Device Deployment)
Python, TensorFlow, FastAPI, Docker

Computer Vision

- Built a lightweight UNet with MobileNetV2 encoder and depthwise decoder achieving 68% Dice coefficient on TuSimple Dataset.
- Quantized and converted model to TensorFlow Lite (27 MB) for Android inference (150 ms).
- Containerized with Docker + FastAPI, integrated Github Actions and unit testing for CI/CD.

LaTeX-OCR: Vision-Language Equation Recognition System
Vision-Language Models (VLMs), NVIDIA NIM API, LLaMA 3.2 Vision, Streamlit, Python

LLM

- Built and deployed a multimodal AI system that converts mathematical equations from images into structured LaTeX code using NVIDIA-hosted LLaMA 3.2 Vision models, enabling production-grade LLM inference via secure API integration.
- Designed an API-driven inference pipeline with prompt structuring, image preprocessing, and secure key management for reliable equation parsing.
- Developed an interactive Streamlit web app with integrated data visualization features for real-time uploads, model inference, and immediate display of prediction results, enhancing user interpretability and deployed it on Streamlit Cloud for public access.

EDUCATION

Bauhaus University
M.S. in Digital Engineering, Machine Intelligence

Weimar, Germany
Oct 2022 – May 2025

- Computer Vision, Machine Learning, Software Engineering, Data Structures and Algorithms, Object Oriented Programming, Optimization, QGIS

IIT Madras
Data Science & Programming (Foundation)

Chennai, India
Jan 2021 – Mar 2022

- Fundamentals of Data Science and Programming, Statistics, Probability

Cochin University of Science and Technology
B.Tech Mechanical Engineering

Ernakulam, India
July 2015 – May 2019

- Engineering Mathematics, Linear Algebra, Calculus

CERTIFICATIONS

- Microsoft Azure AI Essentials Professional Certificate
- Building RAG Agents with LLMs by Nvidia DLI